

VxLEARN Networks

Networking & Cybersecurity Track
Simulated Employment Program

Lab Report: Use Telnet and SSH

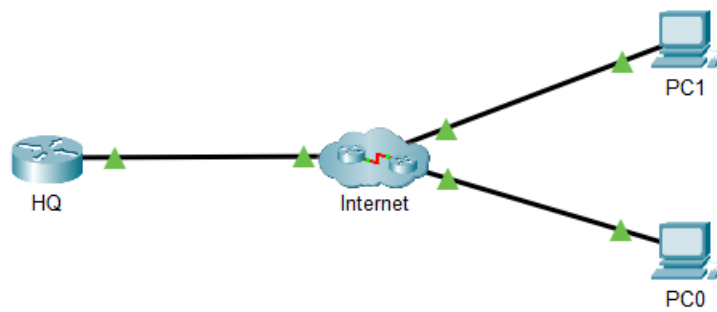
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Addressing Table

Device / Interface	IP Address	Subnet Mask
HQ - G0/0/1	64.100.1.1	255.255.255.0
PC0 - NIC	DHCP	DHCP
PC1 - NIC	DHCP	DHCP



Objectives

- Verify connectivity
- Access a remote device using Telnet and SSH

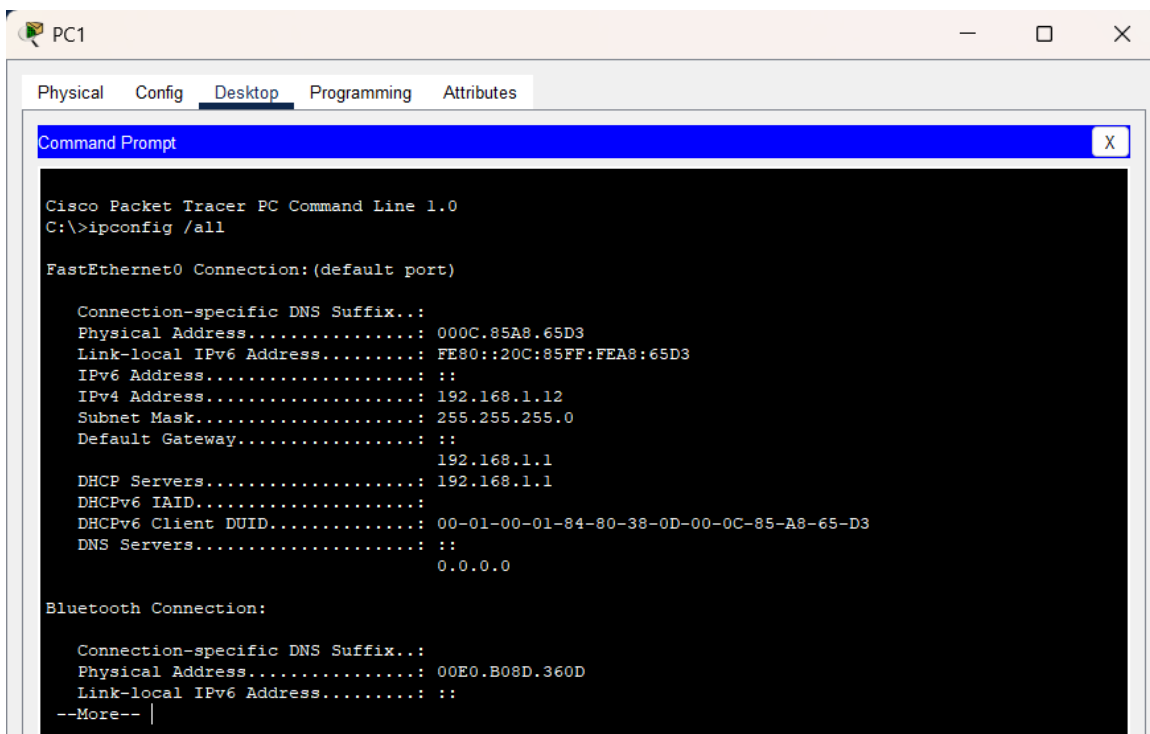
Part 1: Verify Connectivity

Step 1: Verify IP address on a PC

From a PC, click **Desktop**. Click **Command Prompt**.



Command used to verify DHCP IP address: ipconfig

A screenshot of the PC1 Desktop window in Cisco Packet Tracer. The 'Desktop' tab is selected. A 'Command Prompt' window is open, showing the output of the 'ipconfig /all' command. The output displays network configuration details for both FastEthernet0 and Bluetooth connections, including IP addresses, subnet masks, and default gateways.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...: 
    Physical Address. . . . .: 000C.85A8.65D3
    Link-local IPv6 Address . . . . .: FE80::20C:85FF:FEA8:65D3
    IPv6 Address. . . . .: ::
    IPv4 Address. . . . .: 192.168.1.12
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                               192.168.1.1
    DHCP Servers . . . . .: 192.168.1.1
    DHCPv6 IAID. . . . .: 
    DHCPv6 Client DUID. . . . .: 00-01-00-01-84-80-38-0D-00-0C-85-A8-65-D3
    DNS Servers . . . . .: ::
                               0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...: 
    Physical Address. . . . .: 00E0.B08D.360D
    Link-local IPv6 Address . . . . .: ::
    --More-- |
```

PC0 ipconfig

Question:

What command did you use to verify the IP address from DHCP?

Answer:

ipconfig

Step 2: Verify connectivity to HQ

Use the ping command to test reachability to the HQ router:

```
ping 64.100.1.1
```

```
C:\>ping 64.100.1.1

Pinging 64.100.1.1 with 32 bytes of data:

Request timed out.
Reply from 64.100.1.1: bytes=32 time<1ms TTL=253
Reply from 64.100.1.1: bytes=32 time<1ms TTL=253
Reply from 64.100.1.1: bytes=32 time<1ms TTL=253

Ping statistics for 64.100.1.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

Ping HQ

Part 2: Access a Remote Device

Step 1: Telnet to HQ

At the PC command prompt, enter:

```
telnet 64.100.1.1
```

```
C:\>telnet 64.100.1.1
Trying 64.100.1.1 ...Open

[Connection to 64.100.1.1 closed by foreign host]
C:\>|
```

Telnet Attempt

Question:

Were you successful? What was the output?

Answer:

No, the Telnet connection was not successful.
The router is configured to reject insecure Telnet access, so the output typically shows:

- “Password required, but none set” or
- “Connection refused”

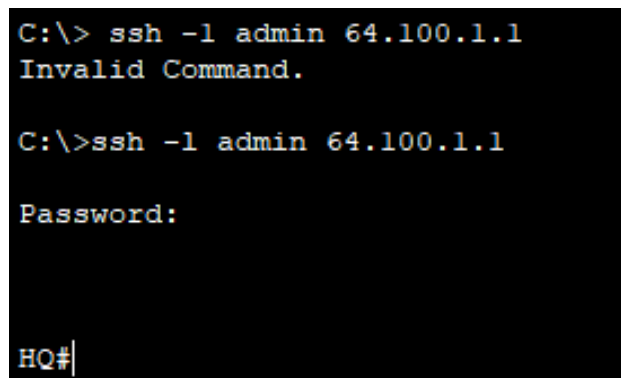
Step 2: SSH to HQ

SSH is configured on the router, so use:

Command: `ssh -l admin 64.100.1.1`

Password: class

Prompt after successful SSH login: HQ#

A terminal window with a black background and white text. The first line shows a command prompt 'C:\>' followed by the command 'ssh -l admin 64.100.1.1' and the output 'Invalid Command.'. The second line shows the same command prompt and command, followed by the output 'Password:'. The third line shows the prompt 'HQ#' with a cursor.

successful SSH session.

Question:

What is the prompt after accessing the router successfully via SSH?

Answer:

After logging in with SSH, the router prompt appears as:

HQ#

This indicates successful login to **privileged EXEC mode**.